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Computer Network Protocols

Chapter Six

The Data Link Layer Protocols

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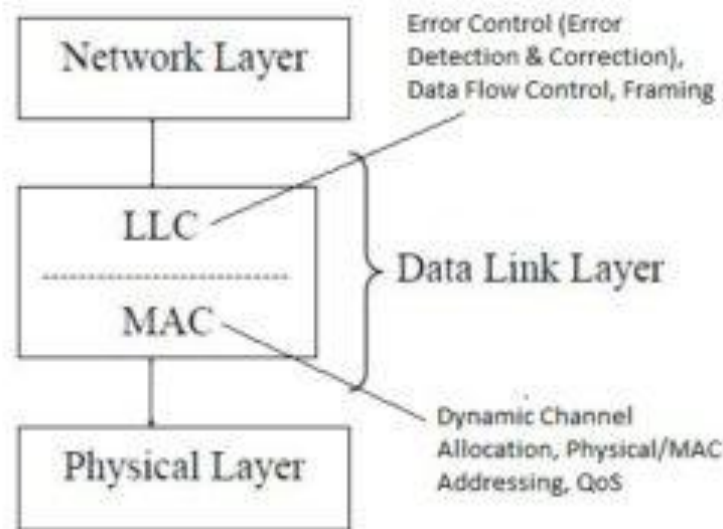
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6.1- Introduction

Data-link layer has the responsibility of transferring datagram from one **node to physically adjacent node** over a link. The data link layer is divided into two sublayers: **Logical Link Control (LLC)** and **Media Access Control (MAC)**. The first, *logical link control, deals with the communication between two adjacent nodes: node-to-node communication. The second function of the data link layer is media access control, or how to share the link.*

Logic link control functions include:

- ***Framing***
- ***Flow Control***
- ***Error control.***



6.2- Framing

Framing in the data link layer separates a message from one source to a destination, by adding a sender address and a destination address. Although the whole message could be packed in one frame; this is not normally done. One reason is that a frame can be very large, making flow and error control very inefficient.

6.3- Flow and Error Control

The most important responsibilities of the data link layer are *flow control* and *error control*. Collectively, these functions are known as **data link control**.

6.3.1- Flow Control

Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before receiving an acknowledgment and is one of the most important duties of the data link layer. The flow of data must **not be allowed to overwhelm the receiver**. Any receiving device has a limited speed at which it can *process* incoming data and a limited amount of memory in which to *store* incoming data. *The receiving device must be able to inform the sending device before those limits are reached and to request that the transmitting device send fewer frames or stop temporarily.* Incoming data must be checked and processed before they can be used. *The rate of such processing is often slower than the rate of transmission.* For this reason, each receiving device has a block of memory, called a **buffer**, reserved for storing incoming data until they are processed. If the buffer begins to fill up, the receiver must be able to tell the sender to halt transmission until it is once again able to receive.

6.3.2- Error Control

Error control is both *error detection* and *error correction*. It allows the receiver to inform the sender of any frames lost or damaged in transmission and coordinates the retransmission of those frames by the sender. In the data link layer, the term error control refers primarily to methods of error detection and retransmission. Error control in the data link layer is often implemented simply. Any time an error is detected in an exchange, specified frames are retransmitted. This process is called ***automatic repeat request (ARQ)***.

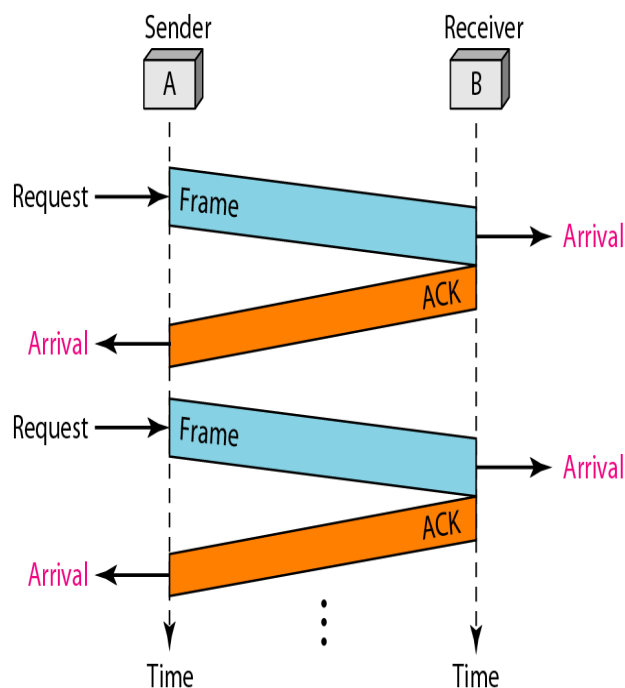
6.4- Protocols

Now let us see how the data link layer can combine framing, flow control, and error control to achieve the delivery of data from one node to another.

6.4.1- Stop-and-Wait Protocol

If data frames arrive at the receiver site *faster than they can be processed*, the frames must be stored until their use. To prevent the receiver from becoming overwhelmed with frames, we need to tell the sender to slow down. There must be feedback from the receiver to the sender.

The protocol is called the **Stop-and-Wait Protocol** because the sender sends one frame, stops until it receives confirmation from the receiver (okay to go ahead), and then sends the next frame.

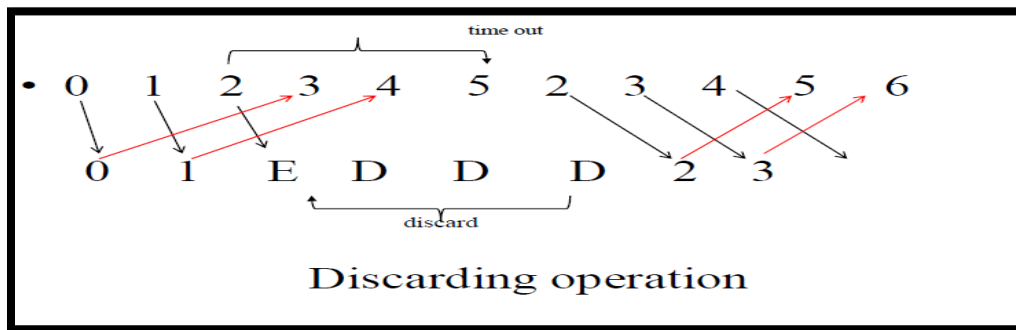


6.4.2- Sliding Window Protocols

To improve the efficiency of transmission, multiple frames must be in transition while waiting for acknowledgment. In other words, we need to let more than one frame be outstanding to keep the channel busy while the sender is waiting for acknowledgment. This can be done by using **Go-back-N** or **Selective-Repeat** protocols rather than Stop-and-Wait. These protocols used for flow and error control but at transport layer.

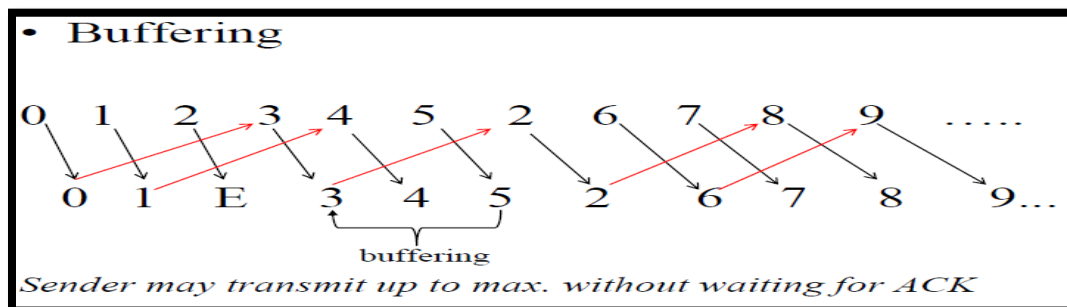
Go-back-N Protocol

If there is one frame k missing, the receiver simply discards all subsequent frames $k+1, k+2, \dots$, sending no acknowledgments. So, the sender will retransmit frames from k onwards. This can be a waste of bandwidth.



Selective-Repeat Protocol

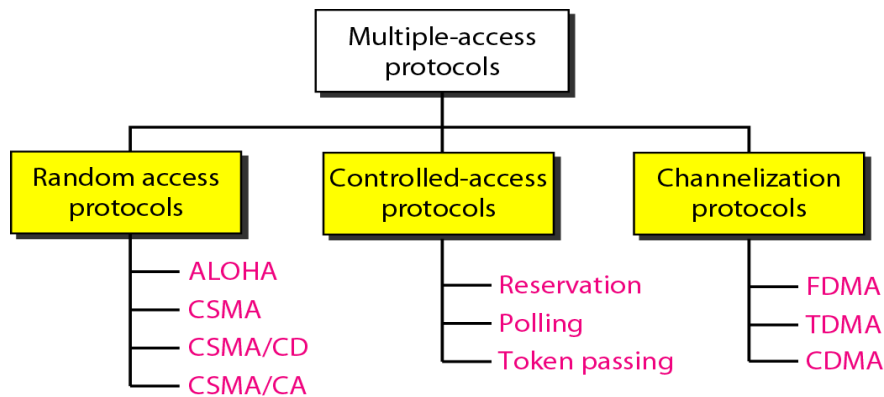
Another strategy is to re-send only the ones that are actually lost or damaged. The receiver buffers all the frames after the lost one. When the sender finally noticed the problem (e.g. no ack for the lost frame is received within time-out limit), the sender retransmits the frame in question.



6.5- Multiple Accesses Protocols

The second sub-layer of the data link layer as we said previously is the media access control. There are two main types of the media access control; **Point-to Point access** and **Multiple Access**.

When there is a *dedicated link can be that can be used by the two devices at any time*; this is known as *Point-to-Point Access*. While, when *the link is shared between pairs of devices that need to use the link*; this is known as a *Multiple Access*.



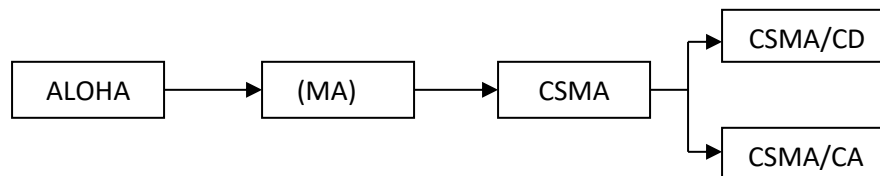
6.5.1- Random Access

In random access, *no station is superior to another station and none is assigned the control over another*. No station permits, or does not permit, another station to send. There is no scheduled time for a station to transmit. Transmission is random among the stations. That is why these methods are called *random access*. In a random-access method, each station has the right to the medium without being controlled by any other station. However, if more than one station tries to send, there is an access conflict-*collision*-and the frames will be either destroyed or modified.

The random-access methods have evolved from a very interesting protocol known as **ALOHA**, which used a very simple procedure called **Multiple Access (MA)**. The method was improved with the addition of a procedure that forces the station to sense

the medium before transmitting. This was called **Carrier Sense Multiple Access (CSMA)**.

This method later evolved into two parallel methods: **Carrier Sense Multiple Access with Collision Detection (CSMA/CD)** and **Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)**. The first tells the station what to do when a collision is detected, while the second tries to avoid the collision.

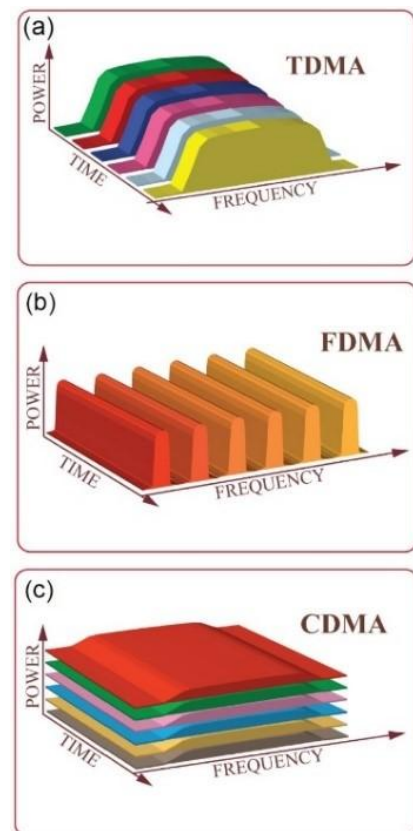


6.5.2- Controlled Access

In controlled access, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations.

6.5.3- Channelization

Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations. Therefore, it is known as FDMA, TDMA, and CDMA. These methods are similar to the multiplexing methods but the main difference is that they occur at the data link layer and used for multiple access for the shared link of the wireless networks.



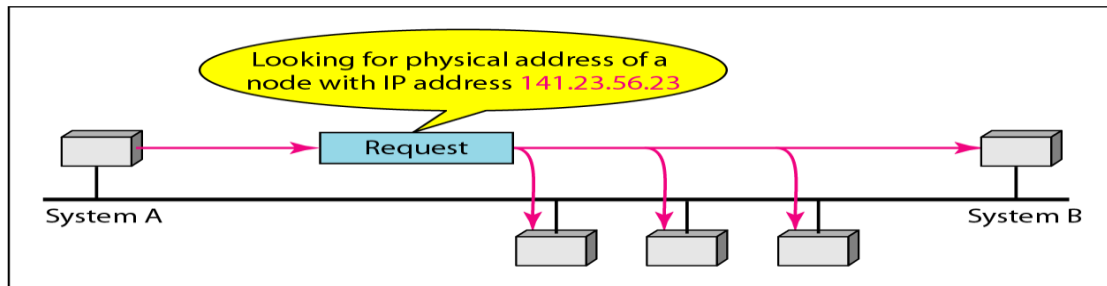
6.6- Layer 2 Protocols

6.6.1- Address Resolution Protocol (ARP)

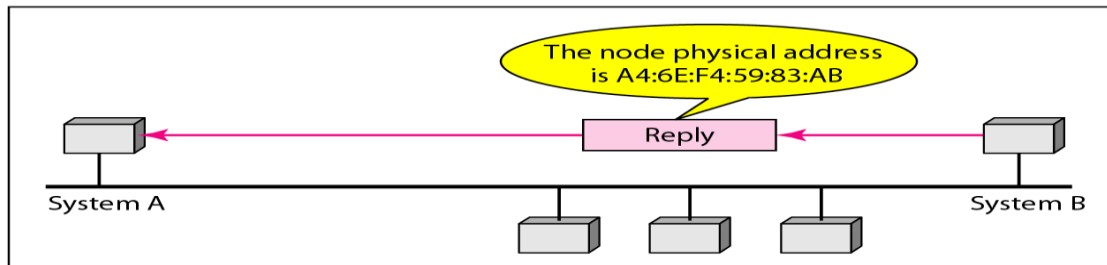
The delivery of a packet to a host or a router requires two levels of addressing: **logical** and **physical**.

- ARP Maps IP addresses to MAC addresses
- ARP **Request** is a broadcast but ARP **reply** is Unicast.
- **ARP tables** contain the MAC and IP addresses of other devices on the network

ARP operation:



a. ARP request is broadcast



b. ARP reply is unicast

6.6.2- Layer 2 Tunneling Protocol (L2TP)

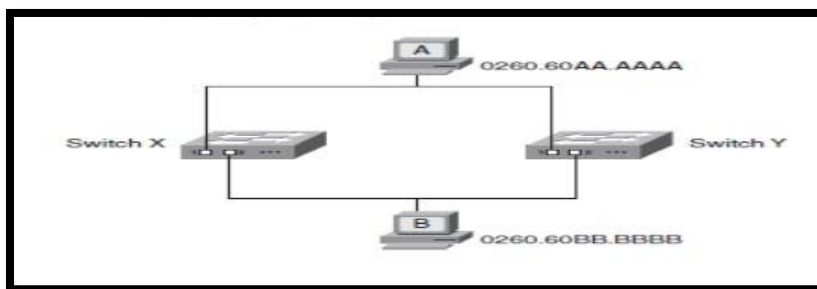
- Is an **extension** of the Point-to-Point Tunneling Protocol (PPTP).
- **Used by** an Internet service provider (ISP) to enable the operation of a **Virtual Private Network (VPN)** over the Internet.
- The goal of a Virtual Private Network (VPN) is to **provide private communications within the public Internet Infrastructure**

6.6.3- Spanning Tree Protocol (STP)

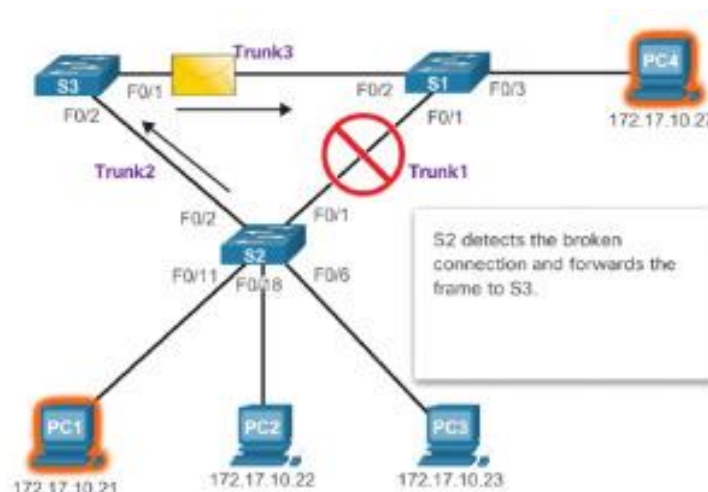
Spanning Tree Protocol (STP) was developed **to prevent the broadcast storms caused by switching loops**, it is a Layer 2 protocol that runs on **bridges and switches**. The main purpose of STP is to ensure that you do not create loops when you have redundant paths in your network.

Switching Loops

Redundancy in a network, such as that shown in Figure below, is **desirable** so that communication can still take place if a link or device fails. **For example**, if switch X in this figure stopped functioning, devices A and B could still communicate through switch Y. However, in a switched network, **redundancy can cause problems which is called switching loop**.



When a switching loop is introduced into the network, a destructive **broadcast storm** will develop **within seconds**. To overcome these problems, you must have a way to logically disable part of the redundant network for regular traffic while maintaining redundancy for the case when an error occurs. Spanning Tree Protocol (STP) does just that.



6.6.4 HDLC Protocol

High-level Data Link Control (**HDLC**) is a bit-oriented layer 2 protocol for communication over point-to-point and multipoint links. It implements the ARQ mechanisms. The primary purpose of **HDLC** is to ensure reliable, error-free, and orderly transmission of data frames between network nodes over a synchronous serial communication link.

6.6.5- Point to Point Protocol (PPP)

Although HDLC is a general protocol that can be used for both point-to-point and multipoint configurations, one of the most common protocols for point-to-point access is the Point-to-Point Protocol (PPP). Today, millions of Internet users who need to connect their home computers to the server of an Internet Service Provider (ISP) use PPP.

Q: Compare between HDLC and PPP protocols

HDLC	PPP
Stand for High-level Data Link Control	Stand for Point-to-Point Protocol
Bit-oriented protocol	Byte-oriented protocol
Support Point to Point and Multipoint Links	Support Point to Point links Only
No Authentication Mechanism	There is Authentication in PPP
Flow Control	No Flow Control
Error Detection and retransmission for recovery	Error detection but there is no recovery mechanism

PPPoE

(Point-to-Point Protocol over Ethernet) is a network protocol that encapsulates PPP frames inside Ethernet frames, commonly used by Internet Service Providers (ISPs) to manage customer broadband connections (DSL, fiber, cable). It enables authentication, encryption, and compression using a username and password to establish a secure session between a home router and the ISP.